American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lecture 03: Multiple Access Techniques I (Data Link Layer I)

OBE Based Questions and Answers

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1. **A university is expanding its campus network where multiple computers share the same communication medium. Explain why the MAC sublayer is required in the Data Link Layer.**

   Answer:

   The Medium Access Control (MAC) sublayer is responsible for controlling access to a shared communication medium.

   Its major responsibilities are:

   - Determining which station can transmit data.
   - Preventing simultaneous transmissions.
   - Reducing transmission collisions.
   - Efficiently utilizing the shared communication channel.

   Without the MAC sublayer, multiple devices could transmit simultaneously, causing collisions and reducing network performance.

2. **A company is designing a communication system where several users must share a single transmission channel. Compare FDMA, TDMA, and CDMA, and recommend the most suitable technique for supporting simultaneous communication.**

   Answer:

Feature	FDMA	TDMA	CDMA
Resource Sharing	Frequency	Time	Code
Simultaneous Transmission	No	No	Yes
Synchronization Required	No	Yes	No
Bandwidth Usage	Fixed	Better	Highest

   CDMA is the most suitable because all users can transmit simultaneously using unique orthogonal codes while sharing the entire frequency spectrum.

3. **A mobile communication provider wants every customer to have a dedicated frequency channel throughout the communication session. Which channelization technique should be used? Explain your answer.**

   Answer:

   The provider should use **Frequency Division Multiple Access (FDMA)**.

   In FDMA:

   - The available bandwidth is divided into frequency bands.
   - Each user receives a dedicated frequency band.
   - The assigned frequency remains reserved during communication.
   - Guard bands prevent interference between adjacent channels.

   Therefore, FDMA is appropriate when continuous dedicated frequency allocation is required.

4. **A satellite communication system requires users to transmit data during specific time intervals. Explain why TDMA is suitable for this application.**

   Answer:

   TDMA divides the communication channel into multiple time slots.

   Each user:

   - Shares the same frequency.
   - Transmits only during its assigned time slot.
   - Waits for the next transmission cycle after completing transmission.

   TDMA efficiently utilizes bandwidth and avoids frequency division, making it suitable for satellite communication systems.

5. **A cellular network allows many users to communicate simultaneously using the same frequency spectrum. Explain how CDMA makes this possible.**

   Answer:

   CDMA allows all users to transmit simultaneously over the entire frequency spectrum.

   Each user is assigned a unique orthogonal code.

   The transmitted signal is

   $$d_1c_1 + d_2c_2 + d_3c_3 + \cdots + d_nc_n$$

   At the receiver, the desired user's code is multiplied with the received signal.

   Because different codes are orthogonal,

   $$c_i \cdot c_j = 0 \qquad (i \neq j)$$

   only the intended user's data is recovered successfully.

6. **Analyze the role of orthogonal codes in CDMA communication.**

   Answer:

   Orthogonal codes ensure that different users do not interfere with each other.

   Their properties are:

   $$c_i \cdot c_j = 0, \quad i \neq j$$

   $$c_i \cdot c_i = N$$

   where N is the code length.

   These properties allow receivers to extract one user's data while eliminating signals from all other users.

7. **A communication engineer needs to generate user codes for a CDMA system. Explain how Hadamard matrices are used for code generation.**

   Answer:

   Hadamard matrices generate mutually orthogonal binary code sequences.

   Each row of the Hadamard matrix represents one user's code.

   For example,

   $$H^1 = [1]$$

   Higher-order matrices such as

$$H^2,\ H^4,\ H^8$$

are generated recursively.

Because every row is orthogonal to the others, simultaneous communication becomes possible without interference.

8. **A network administrator wants to ensure that no station transmits without permission. Which controlled access protocol should be implemented? Justify your answer.**

   Answer:

   Controlled Access protocols should be implemented.

   These include:

   - Reservation
   - Polling
   - Token Passing

   In controlled access, a station must obtain permission before transmitting, preventing collisions and ensuring orderly communication.

9. **A network contains five stations that frequently transmit data. Explain how the Reservation protocol manages channel access efficiently.**

   Answer:

   Reservation divides communication into intervals.

   Each interval begins with a reservation frame containing reservation minislots.

   Each station reserves its minislot before transmitting.

   Only stations that successfully reserve a minislot are allowed to send data during that interval.

   This approach minimizes collisions and improves channel utilization.

10. **A centralized communication network uses one primary device to control all communication. Explain how Polling manages data transmission in this network.**

    Answer:

    Polling consists of one primary station and multiple secondary stations.

    The primary station controls communication.

    It performs two important functions:

    - **Poll**: asks secondary stations whether they have data to transmit.
    - **Select**: instructs a secondary station to prepare for receiving data.

    Only the station permitted by the primary can transmit, thereby avoiding collisions.

11. **A factory network requires fair access so that every device gets an equal opportunity to transmit. Explain how Token Passing achieves this objective.**

    Answer:

    In Token Passing:

    - Stations are arranged in a logical ring.
    - A special packet called a token continuously circulates.
    - Only the station possessing the token can transmit.
    - After transmission, the token is passed to the next station.

    This guarantees fair channel access and prevents transmission collisions.

12. **Evaluate the advantages and limitations of FDMA, TDMA, and CDMA in modern communication systems.**

Answer:

FDMA

Advantages:

- Simple implementation.
- No synchronization required.

Limitations:

- Guard bands waste bandwidth.
- Fixed frequency allocation reduces efficiency.

TDMA

Advantages:

- Better bandwidth utilization.
- No guard bands.

Limitations:

- Requires precise synchronization.

CDMA

Advantages:

- Simultaneous transmission.
- Excellent bandwidth utilization.
- High interference resistance.
- Increased system capacity.

Limitations:

- Complex implementation.
- Requires code management and decoding.

Among the three techniques, CDMA generally provides the highest efficiency for modern wireless communication systems.